ETHRC Motor Controller

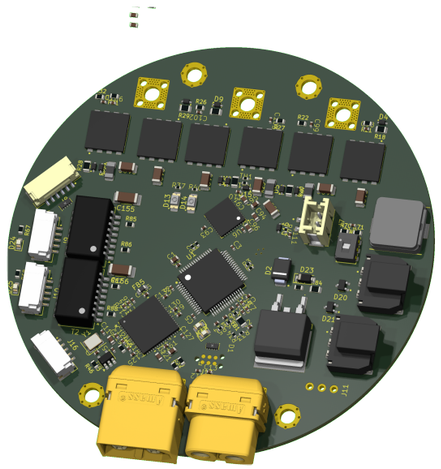
Variant: PRELIMINARY

2026-11-09

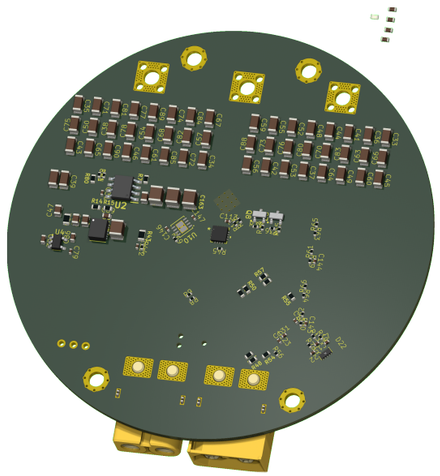
Rev + (Unreleased)

Page	Index	Page	Index	Page	Index	Page	Index
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2	Block Diagram	12	Sensing - Temperature	22	.....	32	.....
3	Project Architecture	13	Sensing - Battery	23	.....	33	.....
4	MCU - Power	14	Sensing - Position	24	.....	34	.....
5	MCU - IOs	15	Comms - EtherCat	25	.....	35	.....
6	Power - Generation	16	Comms - FD-CAN	26	.....	36	.....
7	Power - Connectors	17	Fan Control	27	.....	37	.....
8	Motor Control - Top Level	18	Misc - Holes, Fiducials	28	.....	38	.....
9	Motor Control - Inverter	19	Power - Sequencing	29	.....	39	.....
10	Misc - Board Version, DAC	20	Revision History	30	.....	40	.....

TOP VIEW



BOTTOM VIEW



NOTES

Schematic based off Josh Pieper's moteus controllers.

Not fitted components are marked as ✗

DRAFT - Very early stage of schematic, ignore details.
PRELIMINARY - Close to final schematic.
CHECKED - There shouldn't be any mistakes. Contact the engineer if you find any.
RELEASED - A board with this schematic has been sent to production.

Date: 09-Nov-2026

DESIGN CONSIDERATIONS

DESIGN NOTE: Example text for informational design notes.

DESIGN NOTE: Example text for debug notes.

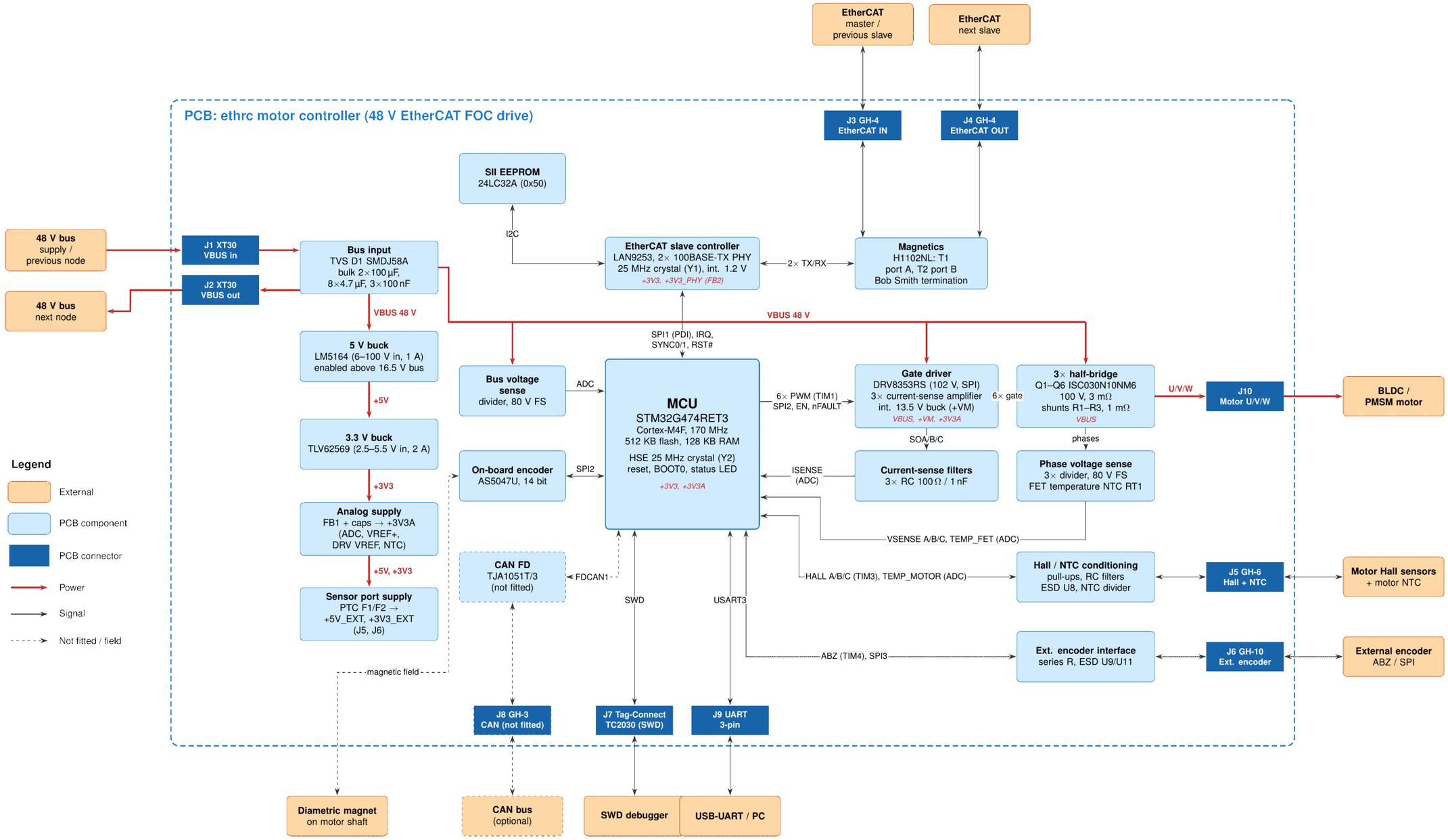
DESIGN NOTE: Example text for cautionary design notes.

DESIGN NOTE: Example text for critical design notes.

LAYOUT NOTE: Example text for critical layout guidelines.

Comments:	Company: ETH Robotics Club		Variant: PRELIMINARY	Git Hash: 1a83c5d
	Board Name: ETHRC Motor Controller		Project Name: BLDC Motor Controller	
Sheet Title: Cover Page	File Name: ethrc_motor_controller.kicad_sch	Designer: Masiar Etemadi	Date: 2024-04-13	Revision: + (Unreleased)
Sheet Path: /		Reviewer:	Size: A3	Sheet: 1 of 20

[2] Block Diagram

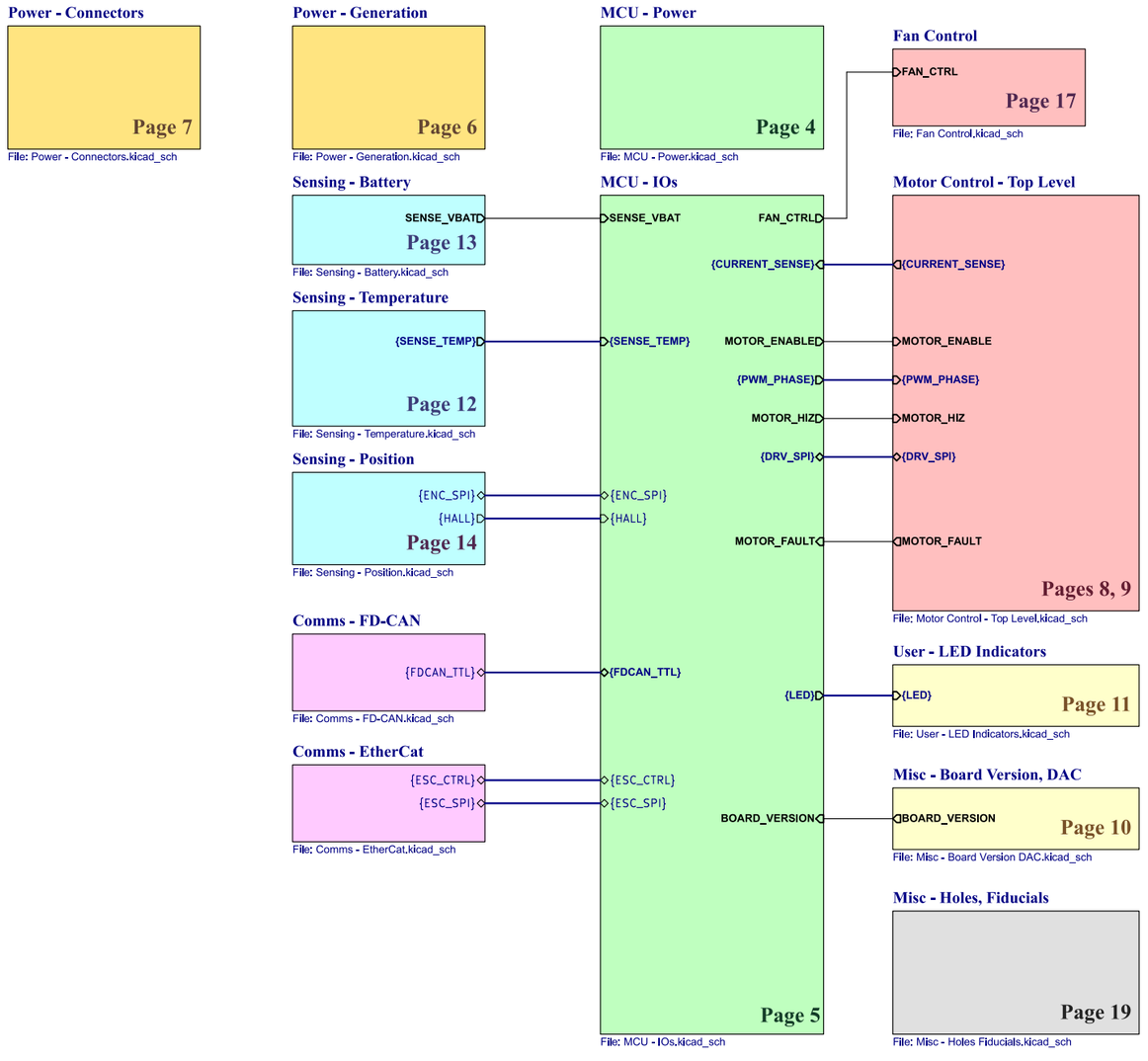


Target specifications:

Input voltage:	36-60 V
Continuous current:	25 A
Peak current:	65 A
Switching frequency:	20-60 kHz

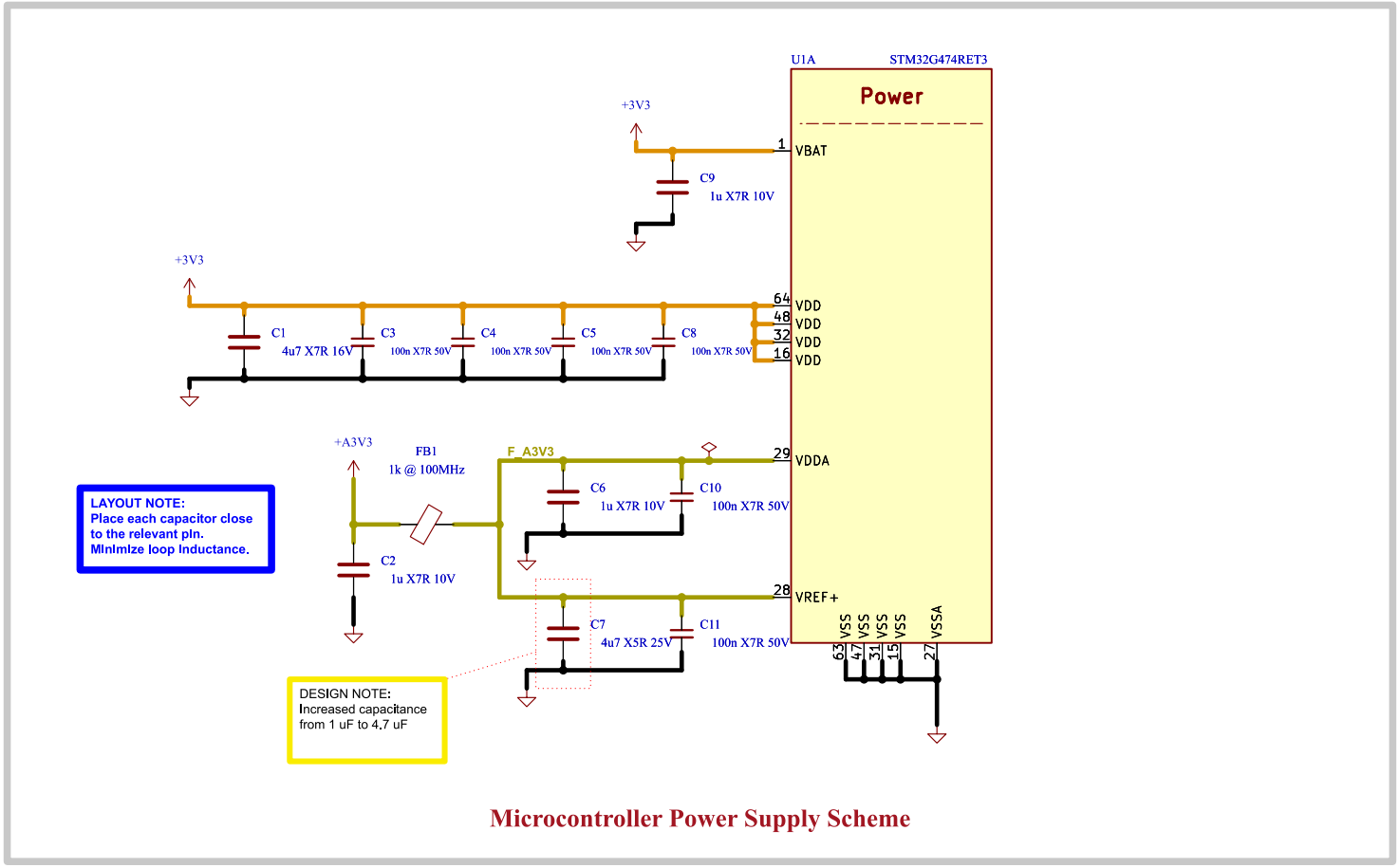
Comments:	Company: ETH Robotics Club		Variant: PRELIMINARY	Git Hash: 1a83c5d
	Board Name: ETHRC Motor Controller		Project Name: BLDC Motor Controller	
Sheet Title: Block Diagram	File Name: Block Diagram.kicad_sch	Designer: Masiar Etemadi	Date: 2024-04-13	Revision: + (Unreleased)
Sheet Path: /Block Diagram/		Reviewer:	Size: A3	Sheet: 2 of 20

[3] Project Architecture



Comments:	Company: **ETH Robotics Club**		Variant: PRELIMINARY	Git Hash: 1a83c5d
	Board Name: **ETHRC Motor Controller**		Project Name: **BLDC Motor Controller**	
Sheet Title: Project Architecture	File Name: Project Architecture.kicad_sch	Designer: Masiar Etemadi	Date: 2023-12-22	Revision: + (Unreleased)
Sheet Path: /Project Architecture/		Reviewer:	Size: **A3**	Sheet: **3** of **20**

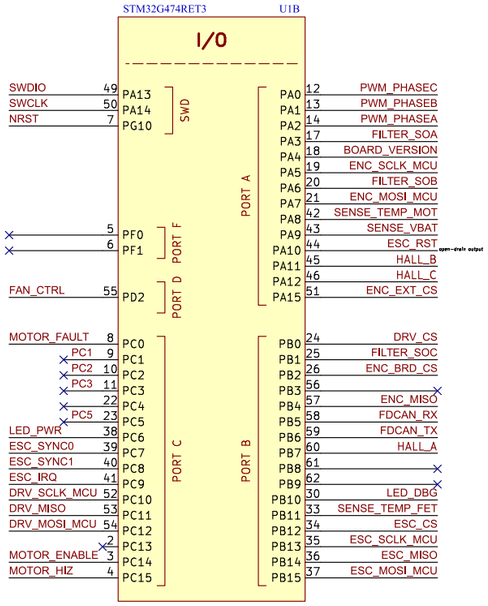
[4] MCU - Power



Microcontroller Power Supply Scheme

Comments: AN5346 STM32G474 Datasheet p.81 J. Pieper ADC investigation	Company: **ETH Robotics Club**		Variant: PRELIMINARY	Git Hash: 1a83c5d
	Board Name: **ETHRC Motor Controller**		Project Name: **BLDC Motor Controller**	
Sheet Title: MCU - Power	File Name: MCU - Power.kicad_sch	Designer: Masiar Etemadi	Date: 2023-12-18	Revision: + (Unreleased)
Sheet Path: /Project Architecture/MCU - Power/		Reviewer:	Size: **A4**	Sheet: **4** of **20**

[5] MCU - I/Os

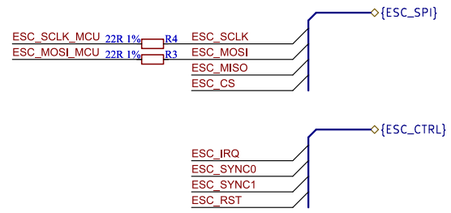


DESIGN NOTE:
We use the LQFP64 version to have extra pins for controlling an external fan.

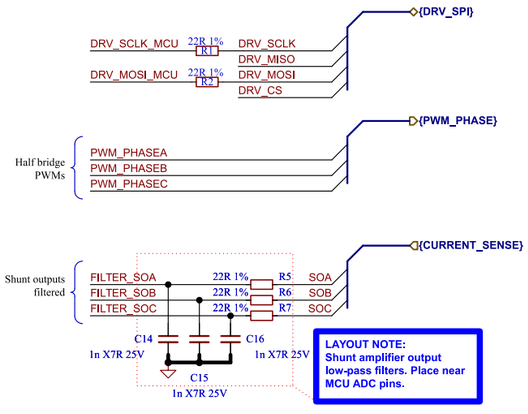
DESIGN NOTE:
Do not forget to flash option bytes when flashing for the first time.

LAYOUT NOTE:
Footprint deviates from datasheet and IPC7351 to allow for soldermask sliver between pads according to PCBWay design rules.

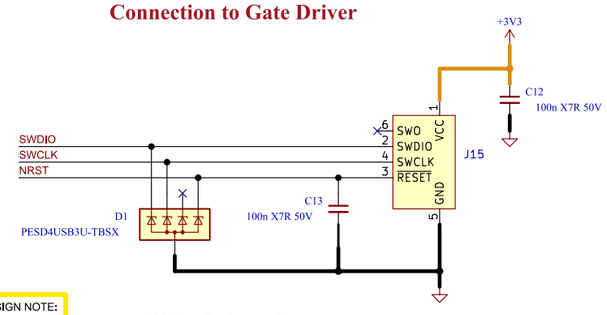
Microcontroller I/O Domain



Connection to EtherCat Subdevice Controller



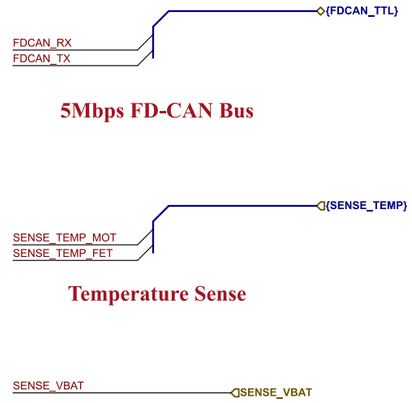
Connection to Gate Driver



DESIGN NOTE:
Ideally TVS should be connected to chassis but makes it harder to route.

DESIGN NOTE:
Only connect GND if connected device does not share ground!

SWD (Debug) Port



5Mbps FD-CAN Bus

Temperature Sense

Battery Voltage Sense

Low-Side Fan Control

Position Sense

Board Version Selection

User LEDs

Gate Driver

MOTOR_ENABLE	TP1	MOTOR_ENABLE
MOTOR_HIZ	TP2	MOTOR_HIZ
MOTOR_FAULT	TP3	MOTOR_FAULT
DRV_SCLK	TP4	DRV_SCLK
DRV_MISO	TP5	DRV_MISO
DRV_MOSI	TP6	DRV_MOSI
DRV_CS	TP7	DRV_CS
PWM_PHASEA	TP8	PWM_PHASEA
PWM_PHASEB	TP9	PWM_PHASEB
PWM_PHASEC	TP10	PWM_PHASEC
SOA	TP11	SOA
SOB	TP12	SOB
SOC	TP13	SOC

Debug

SWDIO	TP14	SWDIO
SWCLK	TP15	SWCLK
NRST	TP16	NRST

ESC_RST	TP17	ESC_RST
ESC_SCLK	TP18	ESC_SCLK
ESC_MISO	TP19	ESC_MISO
ESC_MOSI	TP20	ESC_MOSI
ESC_CS	TP21	ESC_CS
ESC_IRQ	TP22	ESC_IRQ
ESC_SYNC0	TP23	ESC_SYNC0
ESC_SYNC1	TP75	ESC_SYNC1

ENC_SCLK	TP24	ENC_SCLK
ENC_MISO	TP25	ENC_MISO
ENC_MOSI	TP26	ENC_MOSI
ENC_BRD_CS	TP27	ENC_BRD_CS
ENC_EXT_CS	TP28	ENC_EXT_CS

RS-422

LEDs

LED_DBG	TP31	LED_DBG
LED_PWR	TP32	LED_PWR

FD-CAN

FDCAN_RX	TP33	FD_CAN_RX
FDCAN_TX	TP34	FDCAN_TX

Fan

FAN_CTRL	TP35	FAN_CTRL

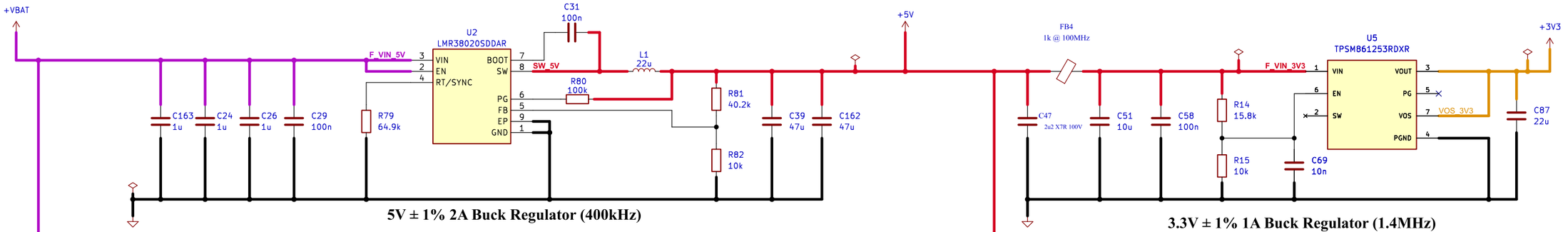
Sense

SENSE_TEMP_MOT	TP36	SENSE_TEMP_MOT
SENSE_TEMP_FET	TP37	SENSE_TEMP_FET
SENSE_VBAT	TP38	SENSE_VBAT

Test Points

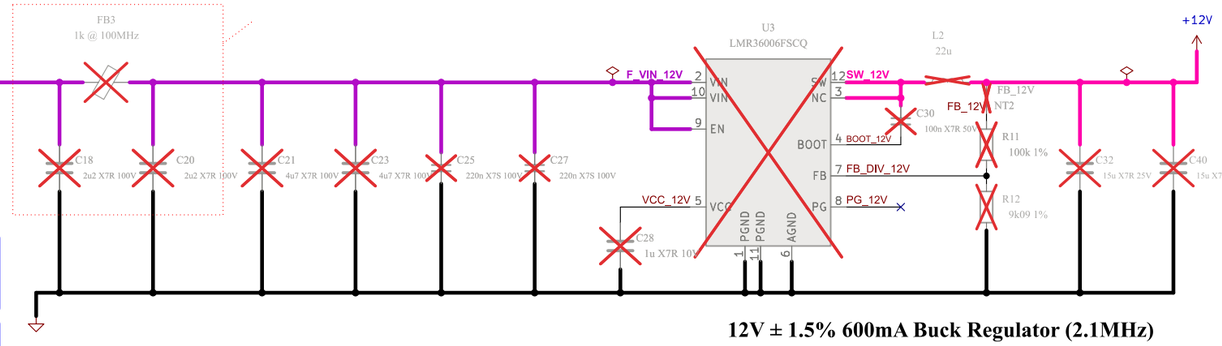
Comments: **References:** Flexible I/O worked examples Flexible I/O source configuration	Company: **ETH Robotics Club**		Variant: PRELIMINARY	Git Hash: 1a83c5d
	Board Name: **ETHRC Motor Controller**		Project Name: **BLDC Motor Controller**	
Sheet Title: MCU - I/Os	File Name: MCU - IOs.kicad_sch	Designer: Masiar Etemadi	Date: 2023-12-20	Revision: + (Unreleased)
Sheet Path: /Project Architecture/MCU - IOs/		Reviewer:	Size: **A3**	Sheet: **5** of **20**

[6] Power - Generation

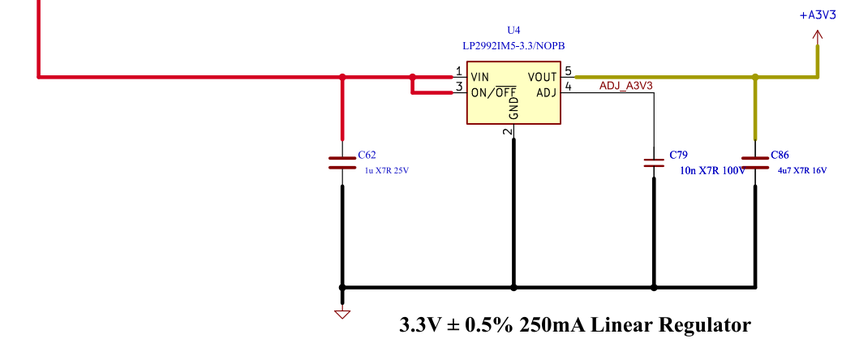


5V ± 1% 2A Buck Regulator (400kHz)

3.3V ± 1% 1A Buck Regulator (1.4MHz)



12V ± 1.5% 600mA Buck Regulator (2.1MHz)



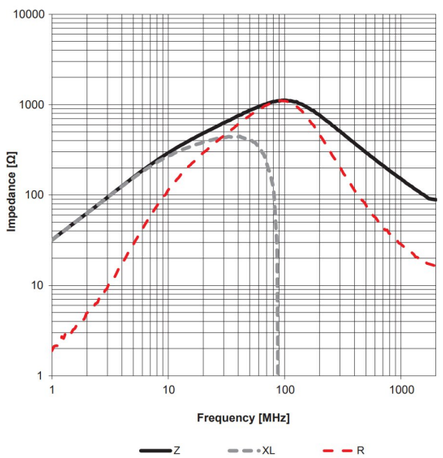
3.3V ± 0.5% 250mA Linear Regulator

LAYOUT NOTE: Route feedback traces away from switching node.

LAYOUT NOTE: Minimize switching node loop areas.

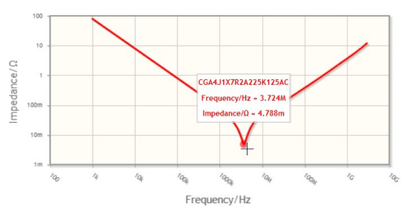
Voltage Regulators

Ferrite bead Impedance vs Frequency curve



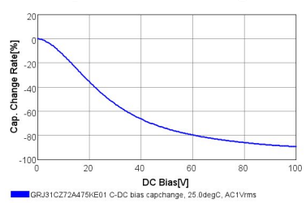
DESIGN NOTE: XL drops around 45th harmonic.

Pi filter capacitor Impedance vs Frequency curve



CGA4J1X7R2A225K125AC

Bulk capacitance Derating curve



DESIGN NOTE: Effective capacitance at 33V is approximately 2.1 uF

Capacitor and Ferrite Bead Parameters

Battery

+VBAT	TP39	+VBAT
+VBAT	TP40	+VBAT
+VBAT	TP41	+VBAT

Filtered Power Rails

F_VIN_12V	TP42	F_VIN_12V
F_VIN_5V	TP43	F_VIN_5V
F_VIN_3V3	TP44	F_VIN_3V3

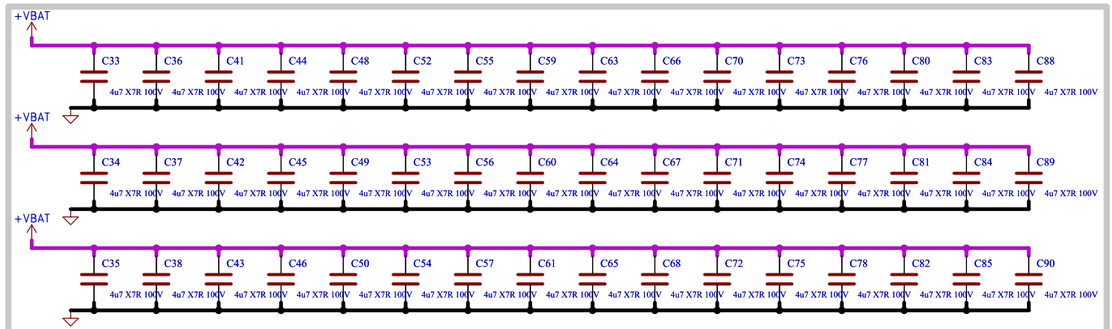
Power Rails

+12V	TP45	+12V
+5V	TP46	+5V
+5V	TP47	+5V
+3V3	TP48	+3V3
+3V3	TP49	+3V3
+3V3	TP50	+3V3
+A3V3	TP51	+A3V3

Ground

GND	TP52	GND
GND	TP53	GND
GND	TP54	GND
GND	TP55	GND
GND	TP56	GND
GND	TP57	GND
GND	TP58	GND

Test Points



DESIGN NOTE: An external precharge circuit is advised to charge the bulk capacitors if multiple boards are connected.

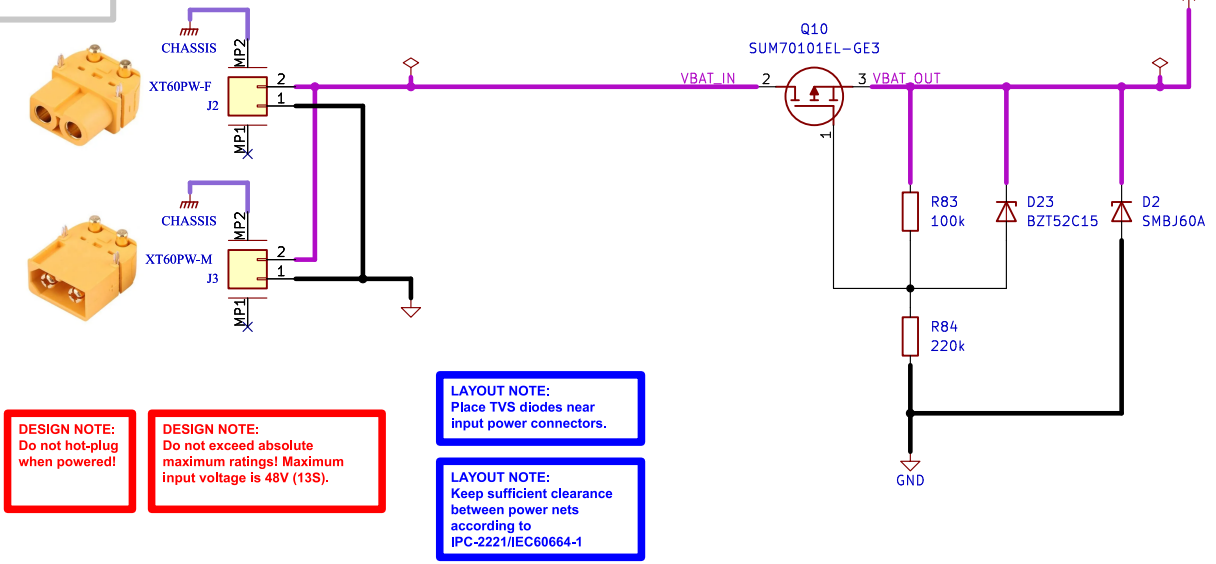
DC-Link Bulk Capacitors. 48 x 4.7 uF = 225.6 uF (100.8 uF at 33V)

DESIGN NOTE: X6S 10uF is marginally better (but worse DC bias derating) at a higher price.

Comments: (+12V (600 mA): Fan power) +5V (600 mA): Gate Driver, LEDs +3V3 (1 A): digital 3.3V +3V3A (250 mA): analog 3.3V	Company: **ETH Robotics Club**	Variant: PRELIMINARY	Git Hash: 1a83c5d
	Board Name: **ETHRC Motor Controller**		Project Name: **BLDC Motor Controller**
Sheet Title: Power - Generation	File Name: Power - Generation.kicad_sch	Designer: Masiar Etemadi	Date: 2024-04-13 / Revision: + (Unreleased)
Sheet Path: /Project Architecture/Power - Generation/		Reviewer:	Size: **A3** / Sheet: **6** of **20**

[7] Power - Connectors

DESIGN NOTE:
It's best to power the controller through the male connector to always have a female connector at the end of the chain.



DESIGN NOTE:
Do not hot-plug when powered!

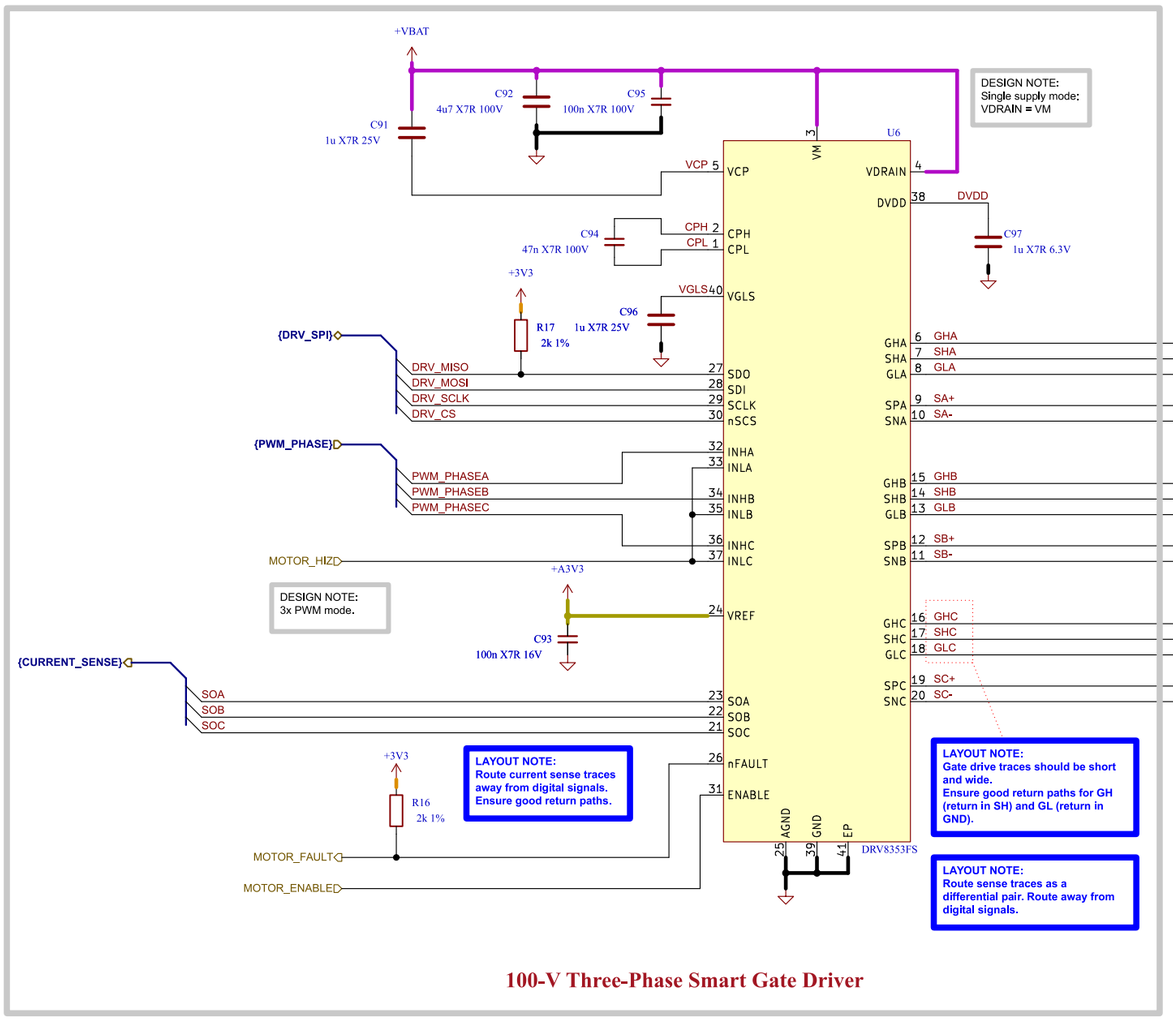
DESIGN NOTE:
Do not exceed absolute maximum ratings! Maximum input voltage is 48V (13S).

LAYOUT NOTE:
Place TVS diodes near input power connectors.

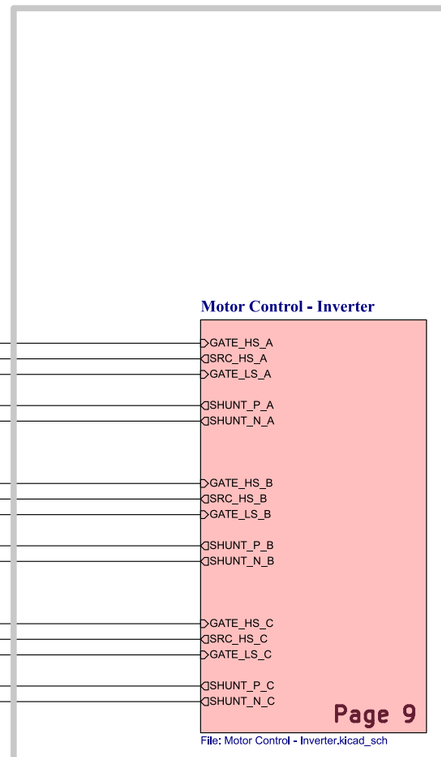
LAYOUT NOTE:
Keep sufficient clearance between power nets according to IPC-2221/IEC60664-1

Comments:	Company: ETH Robotics Club		Variant: PRELIMINARY	Git Hash: 1a83c5d
	Board Name: ETHRC Motor Controller		Project Name: BLDC Motor Controller	
Sheet Title: Power - Connectors	File Name: Power - Connectors.kicad_sch	Designer: Masiar Etemadi	Date: 2023-12-31	Revision: + (Unreleased)
Sheet Path: /Project Architecture/Power - Connectors/		Reviewer:	Size: A4	Sheet: 7 of 20

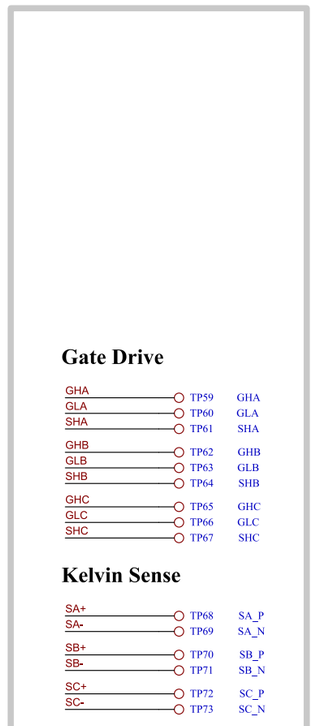
[8] Motor Control - Top Level



100-V Three-Phase Smart Gate Driver



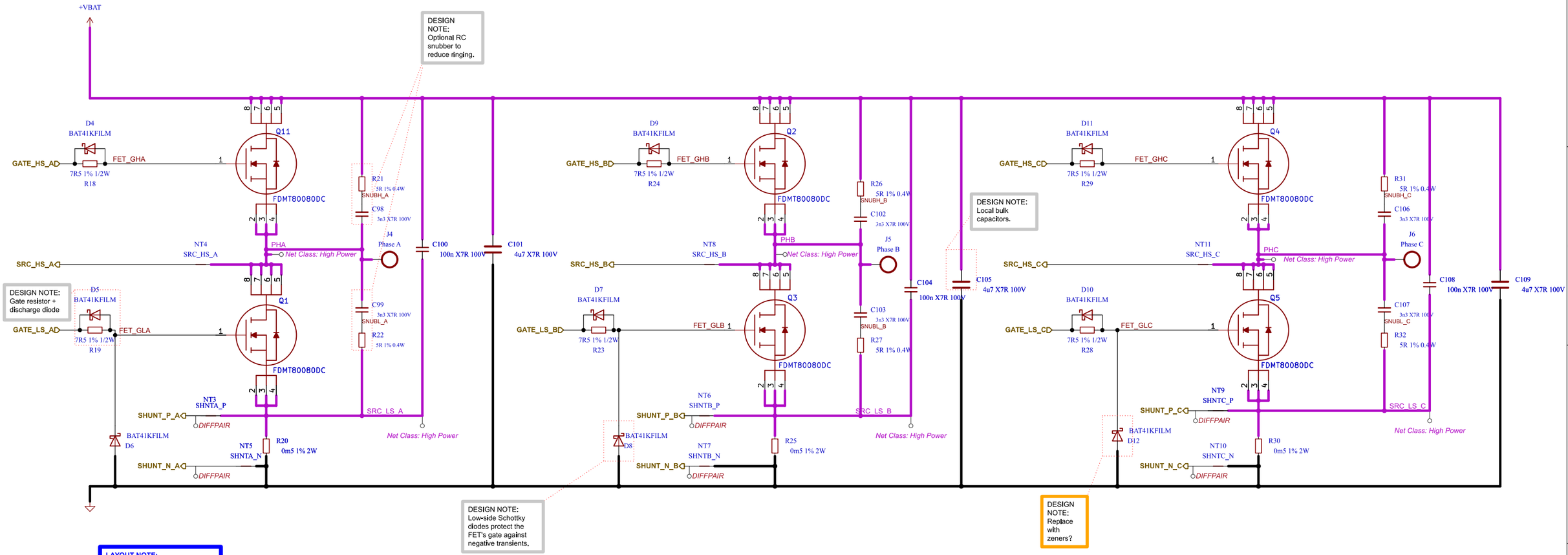
Three-Phase Inverter



Test Points

Comments:	Company: ETH Robotics Club		Variant: PRELIMINARY	Git Hash: 1a83c5d
	Board Name: ETHRC Motor Controller		Project Name: BLDC Motor Controller	
Sheet Title: Motor Control - Top Level	File Name: Motor Control - Top Level.kicad_sch	Designer: Masiar Etemadi	Date: 2023-12-20	Revision: + (Unreleased)
Sheet Path: /Project Architecture/Motor Control - Top Level/		Reviewer:	Size: A3	Sheet: 8 of 20

[9] Motor Control - Inverter



LAYOUT NOTE:
High current traces must be carefully designed. Ensure ground return path does not cross sensitive parts of the board. Use multiple planes for higher current carrying capacity.

LAYOUT NOTE:
Keep sufficient clearance between power nets according to IPC-2221/IEC60664-1.

DESIGN NOTE:
A gate drive current that is too large can damage the FETs!

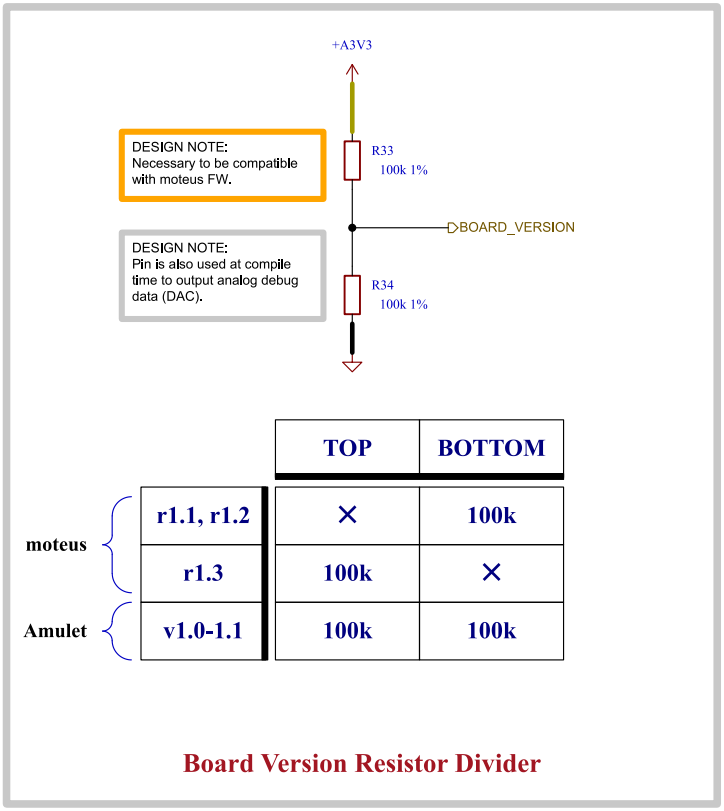
System Design Considerations for High-Power Motor Driver Applications

Best Practices for Board Layout of Motor Drivers

Proper RC Snubber Design for Motor Drivers

Comments:	Company: **ETH Robotics Club**		Variant: PRELIMINARY	Git Hash: 1a83c5d
	Board Name: **ETHRC Motor Controller**		Project Name: **BLDC Motor Controller**	
Sheet Title: Motor Control - Inverter	File Name: Motor Control - Inverter.kicad_sch	Designer: Masiar Etemadi	Date: 2024-01-25	Revision: + (Unreleased)
Sheet Path: /Project Architecture/Motor Control - Top Level/Motor Control - Inverter/		Reviewer:	Size: **A3**	Sheet: **9** of **20**

[10] Misc - Board Version, DAC

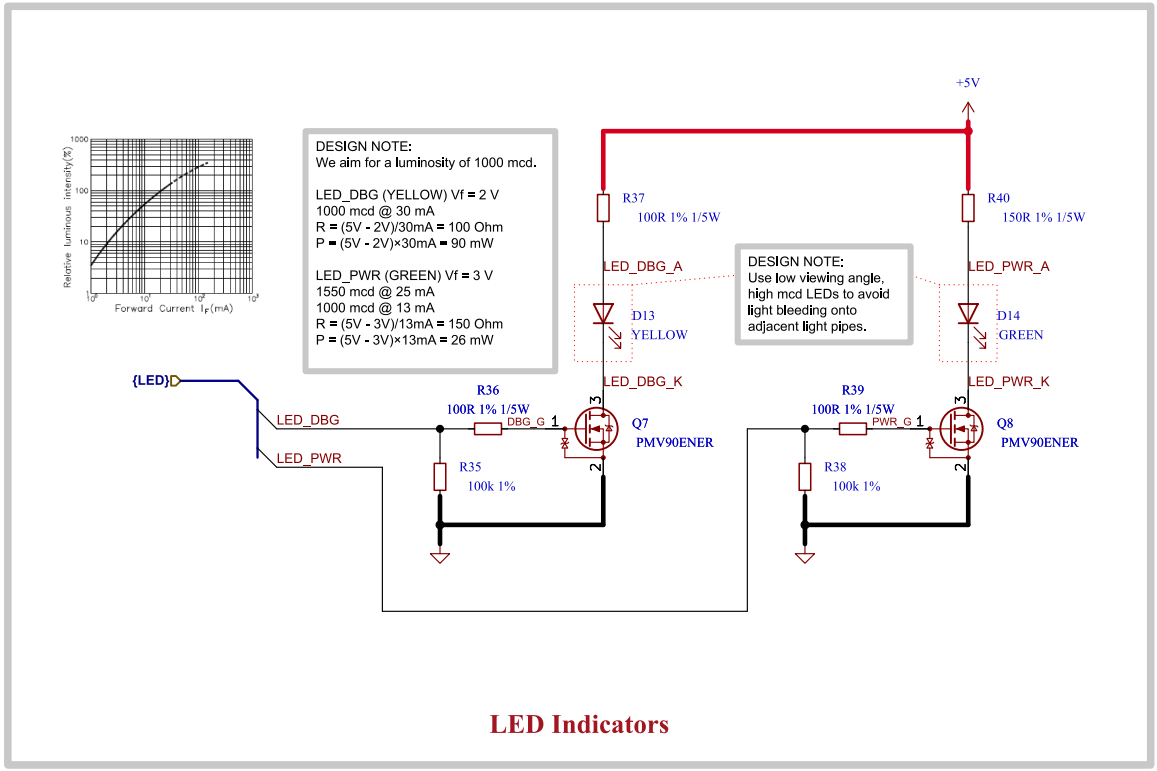


		TOP	BOTTOM
moteus	r1.1, r1.2	×	100k
	r1.3	100k	×
Amulet	v1.0-1.1	100k	100k

Board Version Resistor Divider

Comments:	Company: ETH Robotics Club		Variant: PRELIMINARY	Git Hash: 1a83c5d
	Board Name: ETHRC Motor Controller		Project Name: BLDC Motor Controller	
Sheet Title: Misc - Board Version, DAC	File Name: Misc - Board Version DAC.kicad_sch	Designer: Masiar Etemadi	Date: 2024-04-13	Revision: + (Unreleased)
Sheet Path: /Project Architecture/Misc - Board Version, DAC/		Reviewer:	Size: A4	Sheet: 10 of 20

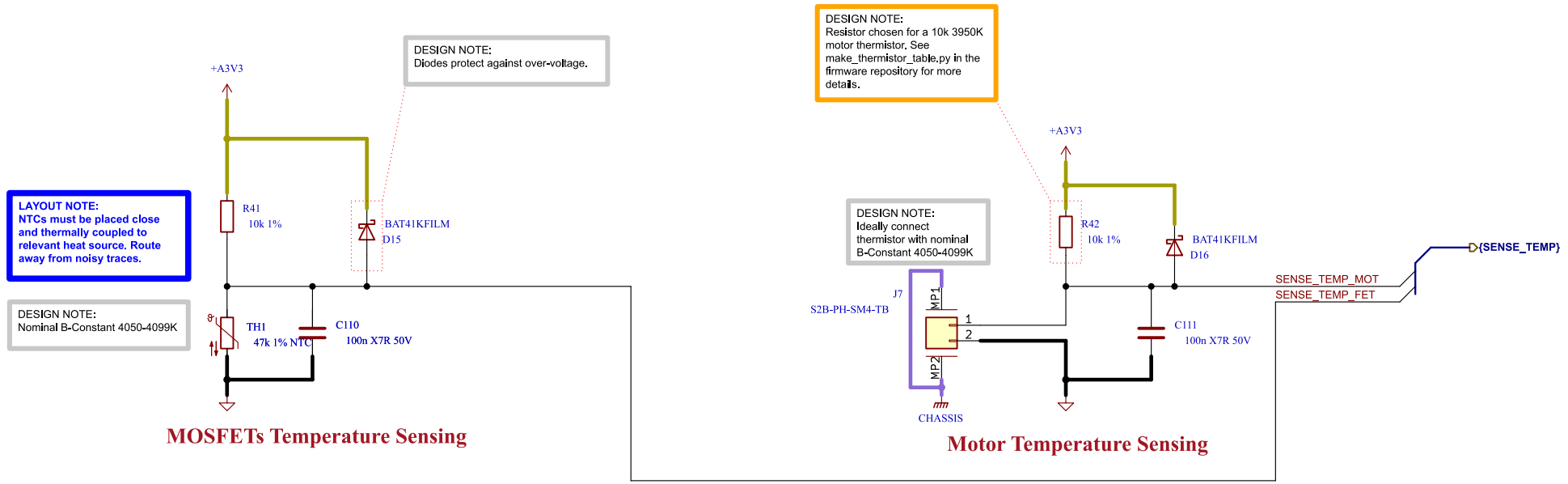
[11] User - LED Indicators



LED Indicators

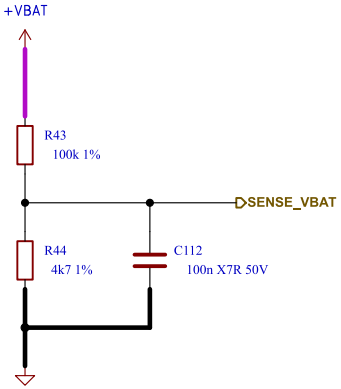
	Comments:	Company: **ETH Robotics Club**	Variant: PRELIMINARY	Git Hash: 1a83c5d
		Board Name: **ETHRC Motor Controller**	Project Name: **BLDC Motor Controller**	
	Sheet Title: User - LED Indicators	File Name: User - LED Indicators.kicad_sch	Designer: Masiar Etemadi	Date: 2023-12-19 / Revision: + (Unreleased)
	Sheet Path: /Project Architecture/User - LED Indicators/		Reviewer:	Size: **A4** / Sheet: **11** of **20**

[12] Sensing - Temperature



Comments:	Company: ETH Robotics Club		Variant: PRELIMINARY	Git Hash: 1a83c5d
	Board Name: ETHRC Motor Controller		Project Name: BLDC Motor Controller	
Sheet Title: Sensing - Temperature	File Name: Sensing - Temperature.kicad_sch	Designer: Masiar Etemadi	Date: 2024-04-13	Revision: + (Unreleased)
Sheet Path: /Project Architecture/Sensing - Temperature/		Reviewer:	Size: A4	Sheet: 12 of 20

[13] Sensing - Battery



DESIGN NOTE:
1/23 temperature stable voltage divider. Use low ppm resistors.

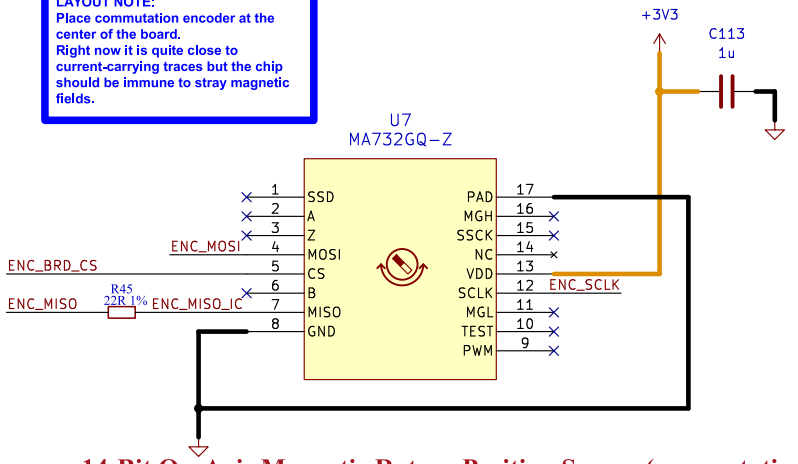
12S: max voltage: 2.19 V
8S: max voltage: 1.95 V
6S: max voltage: 1.46 V

Battery Voltage Sensing

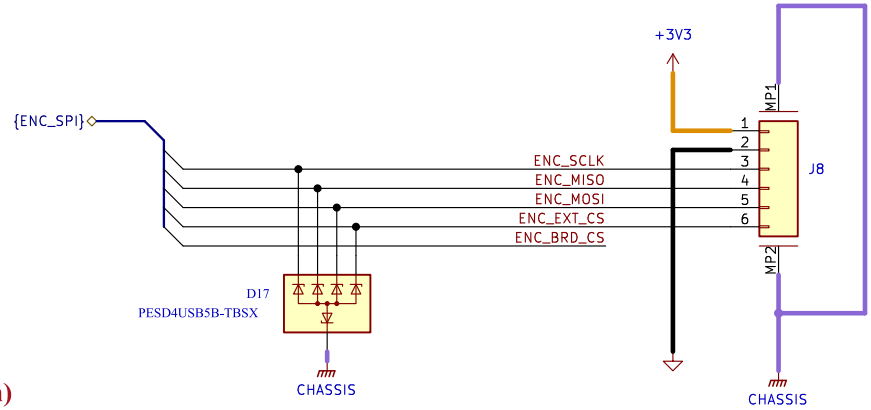
	Comments:	Company: ETH Robotics Club		Variant: PRELIMINARY	Git Hash: 1a83c5d
		Board Name: ETHRC Motor Controller		Project Name: BLDC Motor Controller	
	Sheet Title: Sensing - Battery	File Name: Sensing - Battery.kicad_sch	Designer: Masiar Etemadi	Date: 2023-10-14	Revision: + (Unreleased)
	Sheet Path: /Project Architecture/Sensing - Battery/		Reviewer:	Size: A4	Sheet: 13 of 20

[14] Sensing - Position

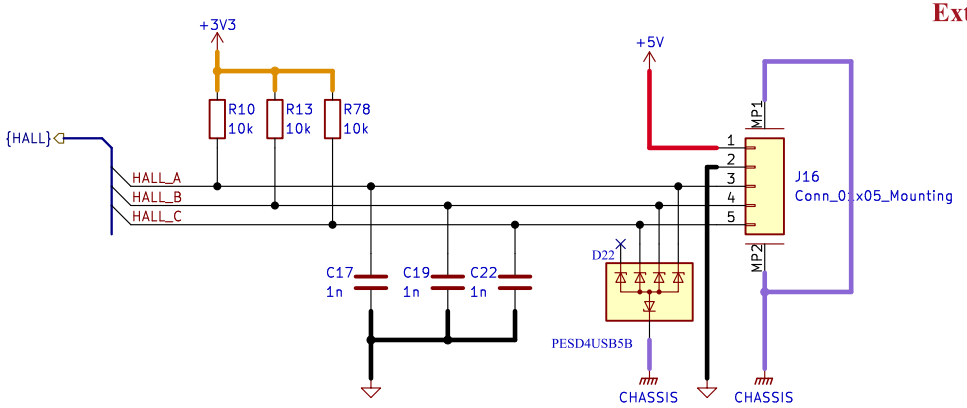
LAYOUT NOTE:
Place commutation encoder at the center of the board.
Right now it is quite close to current-carrying traces but the chip should be immune to stray magnetic fields.



14-Bit On-Axis Magnetic Rotary Position Sensor (commutation)



External 14-Bit On-Axis Magnetic Rotary Position Sensor (disambiguation)



External Hall-Sensor Interface for Commutation

Comments:	Company: ETH Robotics Club		Variant: PRELIMINARY	Git Hash: 1a83c5d
	Board Name: ETHRC Motor Controller		Project Name: BLDC Motor Controller	
Sheet Title: Sensing - Position	File Name: Sensing - Position.kicad_sch	Designer: Masiar Etemadi	Date: 2023-10-14	Revision: + (Unreleased)
Sheet Path: /Project Architecture/Sensing - Position/		Reviewer:	Size: A4	Sheet: 14 of 20

[15] Comms - EtherCat

H1102NL magnetics, Bob Smith (75R + 1n 2kV), JST-GH connectors

ETHERCAT — LAN9253 (SPI PDI, 2 PORTS), Magnetics, SII EEPROM

49.9R terminations to +3V3_PHY (at the LAN9253 pins)

U9 — LAN9253-V/R4X

Mode strap, reset, SPI MISO series

25 MHz crystal

SII EEPROM (I2C address 0x50)

EtherCAT indicators + strap resistors (straps read at reset)

+3V3: VDD33 (5) 1u+100n, VDDIO (14, 20, 32, 37, 47) 100n each, U3 100n

+3V3_PHY: pins 51, 58, 64, T1/T2 centre taps, FB5 output

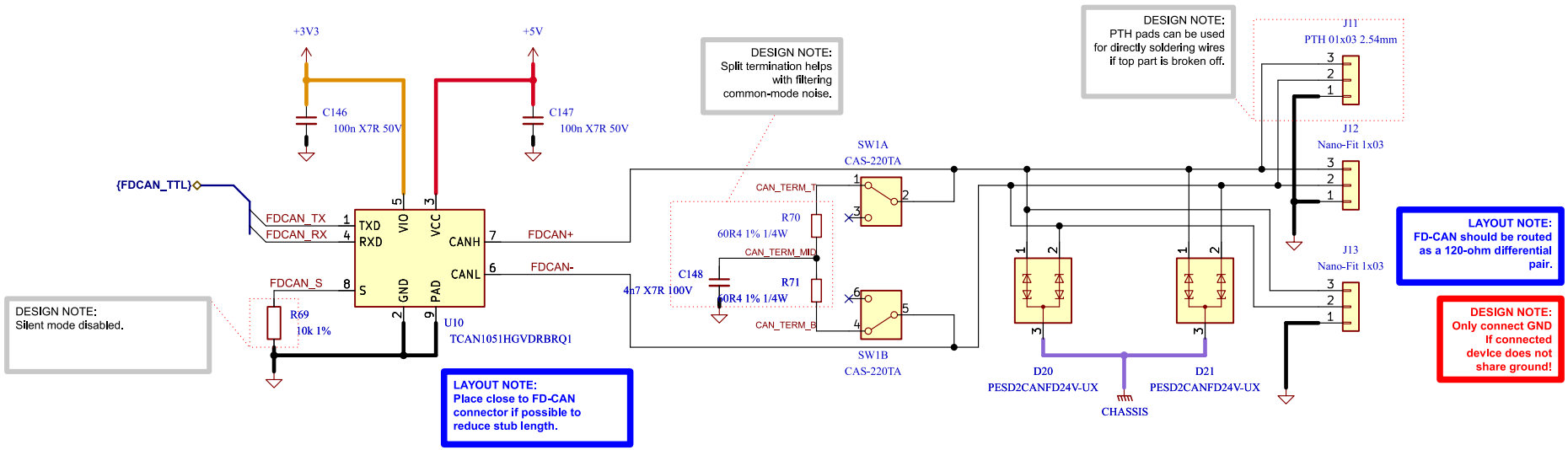
+1V2_ESC (internal regulator): C161 1u low-ESR (DS00003421B 4.1.1) at VDDCR pin 6; 24, 38, 56, 59 100n each

DESIGN NOTE:
WP low: master can write the SII

LAYOUT NOTE: ETHERCAT
* TX/RX pairs 100 ohm differential, matched in-pair, short; 49.9R at the U9 pins.
* Keep PHY side and cable side of T1/T2 apart; no GND plane under the cable side and the GH connectors except the Bob Smith cap return; C155/C156 right at the centre taps, returned to CHASSIS.
* Y1 + caps and R55 (RBIAS) tight to their pins; exposed pad via array to GND.

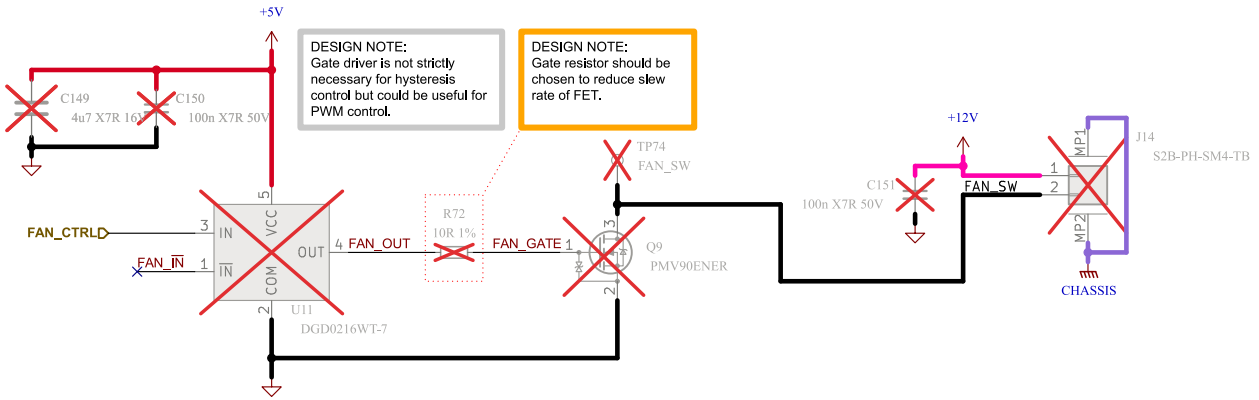
Comments:	Company: ETH Robotics Club		Variant: PRELIMINARY	Git Hash: 1a83c5d
	Board Name: ETHRC Motor Controller		Project Name: BLDC Motor Controller	
Sheet Title: Comms - EtherCat	File Name: Comms - EtherCat.kicad_sch	Designer: Masiar Etemadi	Date: 2025-01-12	Revision: + (Unreleased)
Sheet Path: /Project Architecture/Comms - EtherCat/		Reviewer:	Size: A3	Sheet: 15 of 20

[16] Interface - FD-CAN



Comments:	Company: ETH Robotics Club		Variant: PRELIMINARY	Git Hash: 1a83c5d
	Board Name: **ETHRC Motor Controller**		Project Name: **BLDC Motor Controller**	
Sheet Title: Interface - FD-CAN	File Name: Comms - FD-CAN.kicad_sch	Designer: Masiar Etemadi	Date: 2023-10-15	Revision: + (Unreleased)
Sheet Path: /Project Architecture/Comms - FD-CAN/		Reviewer:	Size: **A4**	Sheet: **16** of **20**

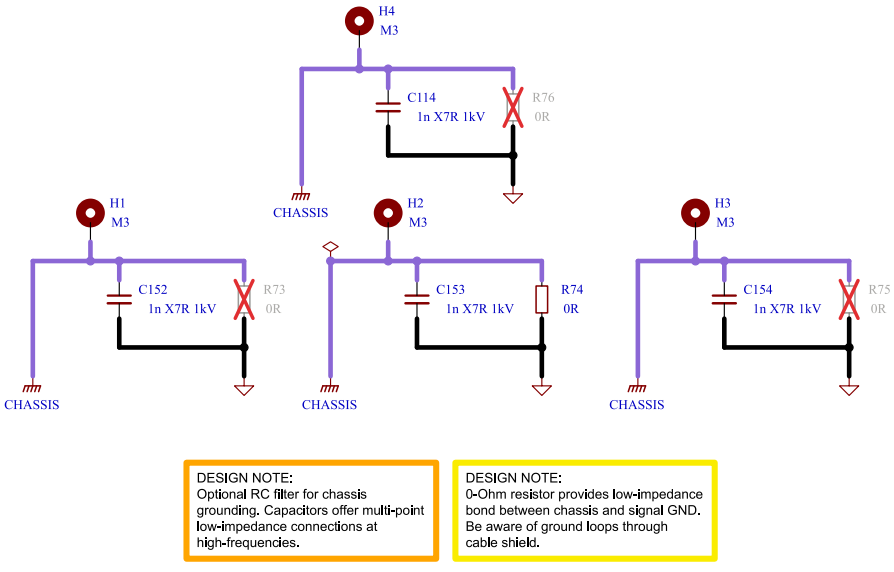
[17] Interface - Fan Control



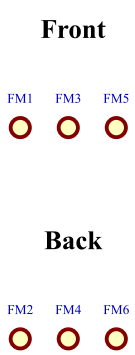
Low-side Fan Switch and Gate Driver

	Comments:	Company: ETH Robotics Club		Variant: PRELIMINARY	Git Hash: 1a83c5d
		Board Name: ETHRC Motor Controller		Project Name: BLDC Motor Controller	
	Sheet Title: Interface - Fan Control	File Name: Fan Control.kicad_sch	Designer: Masiar Etemadi	Date: 2023-11-19	Revision: + (Unreleased)
	Sheet Path: /Project Architecture/Fan Control/		Reviewer:	Size: A4	Sheet: 17 of 20

[19] Misc - Holes, Fiducials



Mounting Holes and Chassis Connection



Fiducial Markers

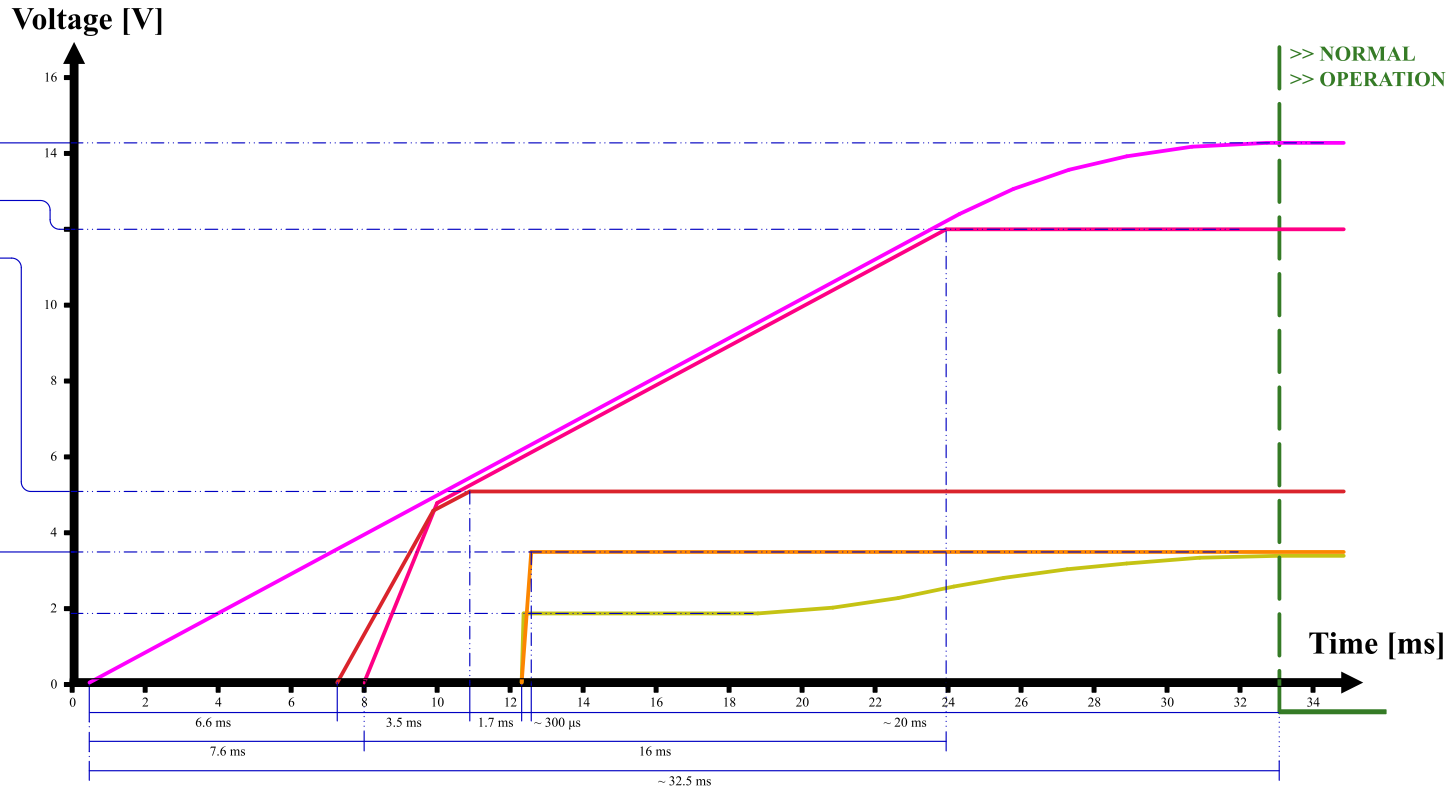
Comments:	Company: ETH Robotics Club		Variant: PRELIMINARY	Git Hash: 1a83c5d
	Board Name: ETHRC Motor Controller		Project Name: BLDC Motor Controller	
Sheet Title: Misc - Holes, Fiducials	File Name: Misc - Holes Fiducials.kicad_sch	Designer: Masiar Etemadi	Date: 2023-10-22	Revision: + (Unreleased)
Sheet Path: /Project Architecture/Misc - Holes, Fiducials/		Reviewer:	Size: A4	Sheet: 19 of 20

[20] Power - Sequencing

NAME	SOURCE	LEVEL
+VBAT	BATTERY	12 - 44V
+12V	LMR36006	12V ± 1.5%
+5V	LMR36506	5V ± 1.5%
+3V3	TPS62172	3.3V ± 3%
+A3V3	LP2992	3.3V ± 0.5%

DESIGN NOTE:
Power sequencing graph measured experimentally.

DESIGN NOTE:
Start-up time of 12V rail depends on +VBAT charging time, which itself depends on the precharge circuit.



Comments:	Company: ETH Robotics Club		Variant: PRELIMINARY	Git Hash: 1a83c5d
	Board Name: ETHRC Motor Controller		Project Name: BLDC Motor Controller	
Sheet Title: Power - Sequencing	File Name: Power - Sequencing.kicad_sch	Designer: Masiar Etemadi	Date: 2024-03-12	Revision: + (Unreleased)
Sheet Path: /Power - Sequencing/		Reviewer:	Size: A4	Sheet: 20 of 20

[21] Revision History

Version 1.0.0 - 2023-12-12

Added

- TVS protection and termination switch to FD-CAN.
- Low-side switched 12V 600 mA source for external fan.
- LDO for analog supply.
- One TVS diode per power connector.
- Second on-board I2C magnetic encoder for disambiguation.
- ESD protection to all interfaces.
- Over-voltage protection on thermistor ADC inputs.
- Pi filters to inputs of buck regulators and MCU analog supply.
- Decoupling caps next to power pins of connectors.

Changed

- CPH-CPL capacitor to 47 nF (gate driver).
- FD-CAN transceiver IC.
- FETs for top cooled variant.
- Input power TVS diode to bidirectional.
- Moved SOx low-pass filter to MCU section.
- PWM_PHASEA with PWM_PHASEC on STM32G474 pinout for easier routing.
- RS422 pinout on connector.
- Buck regulators to optimize for low noise.

Version 1.0.1 - 2024-01-25

Added

- Controller target specifications.
- Credits to moteus on cover page.
- Optional RC-Snubber to power stage.

Fixed

- Chassis guard ring to go around the board.
- CAN and power TVS diodes now go to chassis.
- Clearance between nets to respect IEC60664-1 where possible.
- Comment on precharge.
- Power TVS diode reference designator from "U" to "D".

Changed

- 5V 300 mA buck converter with 600 mA version.
- Chassis-GND capacitor by 1nF 1kV.

Version 1.0.2 - 2024-03-12

Changed

- Power sequencing graph according to experimental data.

Version 1.1.0 - 2024-04-13

Added

- RC snubber passive values.

Fixed

- More vias for VBUS and LMR36006 GND pads.
- Board version voltage reference from +3V3 to +A3V3.

Changed

- Motor thermistor resistor divider to 2kOhm for a 10k 3950K thermistor.

Version 1.1.1 - 2025-01-18

Fixed

- Replace non-stocked components.

Comments:	Company: ETH Robotics Club		Variant: PRELIMINARY	Git Hash: 1a83c5d
	Board Name: ETHRC Motor Controller		Project Name: BLDC Motor Controller	
Sheet Title: Revision History	File Name: Revision History.kicad_sch	Designer: Masiar Etemadi	Date: 2024-01-03	Revision: + (Unreleased)
Sheet Path: /Revision History/		Reviewer:	Size: A4	Sheet: 21 of 20